



USAFA Academic Workload Platform

Executive User Guide & Department Resourcing Architecture

United States Air Force Academy — Academic Division (SINE · SIBS · HASS)

Live Web Dashboard URL

Access Portal: <https://mjr522.github.io/academic-workload-analyzer/>

• Zero installation required • No GitHub account or login needed • Works in Microsoft Edge, Google Chrome, Safari, and Firefox.

100% Air-Gapped Local Privacy (FERPA / CUI Compliant)

All analytical computations, chart rendering, and filtering run **exclusively inside your local browser memory** via the HTML5 FileReader API. **No student records, instructor names, or institutional data are ever uploaded, transmitted, or stored on any external server.**



Leadership Quick-Start Guide (How to View Your Data)

1. **Open the Website:** Navigate to <https://mjr522.github.io/academic-workload-analyzer/>.
2. **Load the Emailed Data File:**
 - Drag & drop the attached `workload_data.json` file anywhere onto the browser window, **or**
 - Click the " **Load Local Data JSON**" button in the top-right header and select the file.
3. **Instant Exploration:** The dashboard will immediately populate all metrics, charts, and tables across all three USAFA academic schools.



Interactive Dashboard Capabilities

1. Executive Overview Pane

- **Academic School Scope Selector:** Filter high-level KPIs across the entire **USAFA Academic Division** (all 21 departments) or isolate by school: **SINE** (Engineering), **SIBS** (Basic Sciences), or **HASS** (Humanities & Social Sciences).
- **2x2 Resourcing Matrix (Quadrant Scatter Plot):**
 - **X-Axis (Course Prep Load):** Weighted sections taught per instructor.
 - **Y-Axis (Student Contact Load):** Total cadet seats taught per instructor.
 - **Bubble Size:** Total student volume generated by the unit.
 - *Note:* Core Engineering (`ESIS`) is filtered from the bubble chart to prevent skewing benchmark averages.
- **Student Credit Hours (SCH) Ranking:** Benchmarks total academic credit volume delivered across units.
- **Small Section Monitoring:** Tracks elective and seminar sections with ≤ 10 cadets across departments.

2. Department Drilldown Pane

- **Comprehensive Footprint:** Real-time counts of unique catalog courses, active classroom sections, cadet seats, total SCH, and declared majors.
- **Cadet Major Pipeline:** Stacked class-year breakdown tracking students from Fourth-Class (Freshman) to First-Class (Senior) years.
- **Billet & Staffing Health:** Visualizes authorized vs. vacant billets, military vs. civilian filled lines, and courtesy adjuncts.
- **Academic Advising Metrics:** Total institutional advisees, active advisors, and average advisee burden per faculty member.
- **Department Assigned Faculty Table:**
 - Deduplicated roster of faculty assigned to the department with contact loads, average class sizes, and courses taught.
 - **# of Advisees Column:** Direct attribution of cadet advisees per faculty member.
 - **Two-Way Interactive Sorting:** Click any column header once to sort highest-to-lowest; click again to reverse.
- **Department Course Offerings & Sections Table (New):**
 - Granular audit of every course offering, class section code (e.g. **M1A** , **T2B**), term, enrolled class size, and credit units.
 - **Real-Time Search:** Filter courses, titles, sections, or instructor names instantly.
 - **Clickable Profiles:** Click any instructor name to view their complete Workload Profile modal.
 - **Enrollment Badges:** Identifies sections with ≤ 10 cadets in amber and senior capstone design courses.

3. Curriculum & Capstones Pane

- Audits small section proliferation across all course catalog levels and tracks multi-section senior design capstone fragmentation.

4. Faculty Workload Directory & Individual Modals

- Campus-wide deduplicated directory showing actual vs. expected teaching sections, delta indicators, contact load, and advisee counts.
- Clicking any instructor opens an individual modal showing every assigned course section, co-instructors, and cadet count.

5. "What-If" Staffing Scenario Sandbox

- Interactive staffing simulator: model the impact of new hires, retirements, role reassignments, and department transfers.
- Click " ⚡ **Recalculate Scenario Live**" to immediately see projected changes in teaching loads and contact hours.
- Export scenario proposal packages via " 📄 **Export Scenario JSON**".

6. Privacy Name Masking (Briefing Mode)

- Click " 👁 **Names Visible** / 🔒 **Names Masked**" in the top header to instantly replace instructor names with anonymous identifiers (e.g. `Faculty 01` , `Faculty 02`) for public or wide-audience presentations.

USAFA Institutional Structure & Academic Mapping

The workload analysis engine organizes the entire USAFA Academic Division into its three official schools and 21 departments:

School	Dept Code	Official Department Name	Included Disciplines & Subjects
SINE	ESME	Mechanical Engineering	MECHENGR , SYSENGR
SINE	ESCS	Computer Science	COMPSCI , CYBERSCI
SINE	ESAN	Aeronautics	AEROENGR
SINE	ESCE	Civil & Environmental Engineering	CIVENGR , CIVENG
SINE	ESECE	Electrical & Computer Engineering	ECE
SINE	ESAS	Astronautics	ASTRENGR , SPACE
SINE	ESIS	SINE Core Engineering (<i>Interdisciplinary</i>)	ENGR (<i>Cross-department taught</i>)
SIBS	BSBI	Biology	BIOLOGY
SIBS	BSCH	Chemistry	CHEM
SIBS	BSMS	Mathematical Sciences	MATH , DATASCI , OPSRSCH
SIBS	BSPM	Physics & Meteorology	PHYSICS , METEOR
HASS	HSBL	Behavioral Sciences & Leadership	BEHSCI , LDRSHP
HASS	HSEG	Economics & Geosciences	ECON , GEO
HASS	HSEN	English & Fine Arts	ENGLISH , CREATART , COMMSTRT
HASS	HSHI	History	HISTORY
HASS	HSLA	Law	LAW
HASS	HSLC	Foreign Languages & Cultures	SPANISH , FRENCH , GERMAN , etc.
HASS	HSMA	Management	MGT
HASS	HSMI	Military & Strategic Studies	MSS
HASS	HSPS	Political Science	POLSCI , SOCSCI
HASS	HSPY	Philosophy	PHILOS

Note on Commissioning Education (CE): Non-academic commissioning education courses (CE) are excluded from this analysis as they are not administered by the Academic Division.

⚙️ Analytical Attribution Formulas

- **Canonical Instructor Resolution:** Advanced regex normalization removes suffixes (Jr. , Sr. , II , III), middle names, and initials to merge disparate campus records into verified canonical faculty lines.
- **Co-Teaching Split (\$1/k\$ Model):**

$$\text{Section Share} = (1 / k) \times \text{Duration Weight}$$

$$\text{Cadet Contact Load} = (\text{Enrolled Cadets} / k) \times \text{Duration Weight}$$

- **Duration Weighting:** Full-semester courses = **1.0** ; quarter/half-semester courses = **0.5** .
- **Student Credit Hours (SCH):** $SCH = \text{Enrolled Cadets} \times \text{Credit Units}$.
- **Cadet Advising Load:** Aggregates institutional cadet records, divides co-advising duties, and maps advisee counts directly to teaching faculty profiles.